Irish physicist **John Bell** (1928–90) devised an experiment to prove the nonlocal nature of the world. His work showed that it would be possible, in principle, to make measurements on two quantum entities (such as photons) that had once been in contact, and discover if they remained "aware" of each other when far apart. Such experiments were actually carried out by Alain Aspect in Paris, and proved conclusively that quantum systems are nonlocal.

experiments, an atom is induced to emit two photons simultaneously in opposite directions. The common origin of these photons means that they are correlated with one another and, according to the equations of quantum physics, they remain "entangled" even when they are far apart—as if they formed a single particle. The experiments proved that measuring the properties of one of the photons on one side of the lab affected the other photon on the other side of the lab instantaneously. Through these experiments, nonlocality could be seen at work.

By the middle of the 1990s, researchers in Geneva had extended these experiments and sent photons along fiber-optic cable 6.2 miles (10 km) in length. The experiments still show nonlocality. The two photons acted like one particle even when they were more than six miles apart. And it cannot be over-emphasized that these are genuine experimental results, not "merely" the predictions of the theory.

conveying coded messages

But although some influence links the two photons instantaneously, no useful information travels between them faster than the speed of light. The measurement of photon A disturbs photon B in a random fashion. Observations can show that photon B has been disturbed, indicating that something happened to photon A, but cannot reveal precisely what it was that happened to photon A. If, however, some (incomplete) information

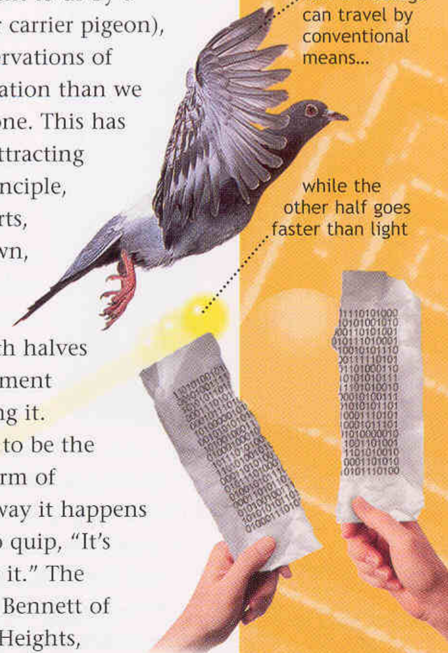
about what was done to photon A is sent to us by a conventional route (by letter, email, or carrier pigeon), we can put this together with our observations of photon B to end up with more information than we would get by one of the two routes alone. This has implications for cryptography and is attracting considerable funding as a result. In principle, a secret message can be sent in two parts, neither of which makes sense on its own, but one of which involves quantum entanglement. The whole message travels no faster than light because both halves are needed, but the quantum entanglement cannot be intercepted without changing it.

It is even possible for the "message" to be the original particle, photon A, itself—a form of teleportation, although not quite the way it happens in *Star Trek*, leading many physicists to quip, "It's teleportation, Jim, but not as we know it." The possibility was pointed out by Charles Bennett of the IBM Research Center in Yorktown Heights, New York, in a paper published in 1993. Since then, experiments have confirmed that the idea works, at least on the laboratory scale.

more than a copy

The idea depends on the fact that if one entity is indistinguishable from another entity in every way, then it *is* that entity. When you make a photocopy, you have two "entities" and you know which is the original. But if the original were destroyed in the process, and the "copy" was identical in every way to the original, then it would not really be a copy at all. It would *be* the original.

Quantum teleportation works like this. Two entangled photons are prepared in the usual way. One is carried off to a distant place (the Moon, for example). The other one



a message in two halves

Quantum nonlocality means that when one "entity" is tweaked, another instantly responds, no matter how far away it is. To understand that response—and its meaning—you have to wait for further information to arrive by a conventional route.